EXPERIMENT – 2

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TITLE:

Design a UI to evaluate the effect of chunking on user memory using a Memory Recall Task

AIM

To design and analyze a user interface that applies the **chunking principle** to improve short-term memory recall and evaluate how grouping visual elements affects user performance.

DESCRIPTION OF THE TASK

The Memory Recall Task consists of multiple screens where users:

1. Read instructions
2. Observe grouped visual items for a limited time
3. Recall the items from a new set
4. Receive feedback based on recall accuracy

The experiment focuses on how **chunking visual information** helps users remember more items efficiently.

FRAME 1: INSTRUCTION PAGE

Purpose of the Screen

- Introduces the task to the user.
- Sets expectations about time limits and recall behavior.
- Prepares users mentally for the memory challenge.

UI Elements Used

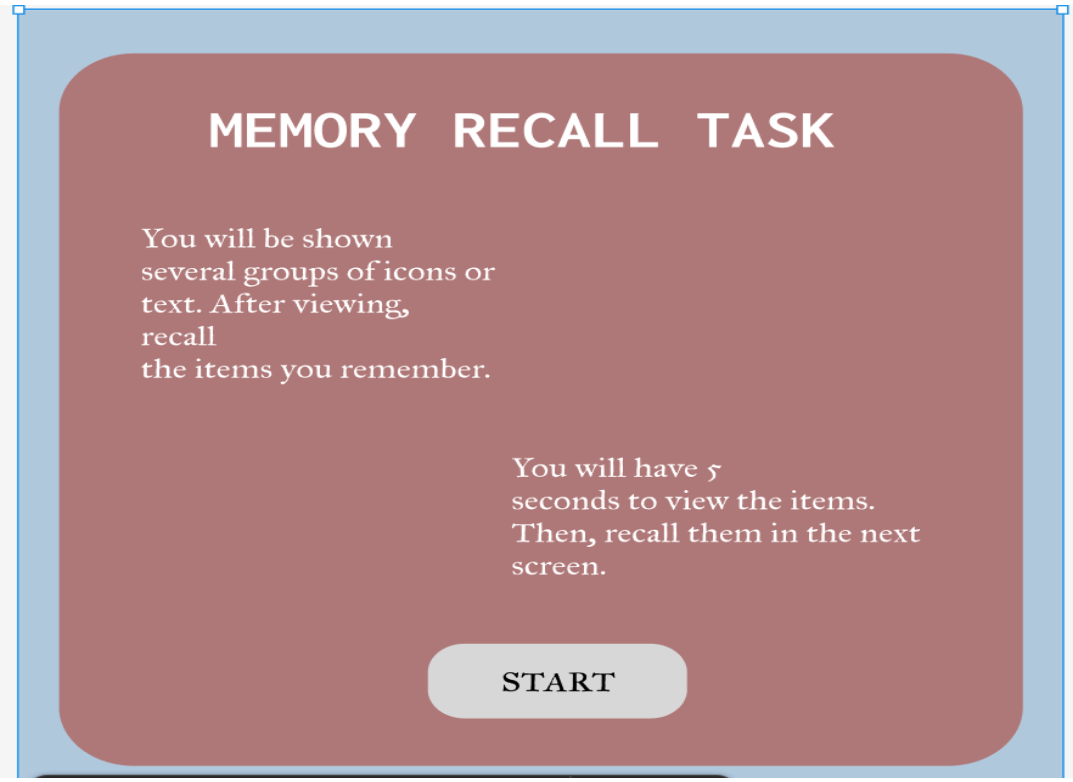
- **Title:** “MEMORY RECALL TASK” – clearly communicates the task.
- **Instruction Text:** Short paragraphs explaining what the user will see and do.
- **Time Information:** Mentions the 5-second viewing duration.
- **START Button:** Indicates the beginning of the task.

Chunking Analysis

- Instructions are **broken into small, readable chunks**, reducing cognitive load.
- Related information is grouped logically (task description + time constraint).
- Large spacing and minimal text prevent overwhelming the user.

UX & Cognitive Benefits

- Improves comprehension before starting the task.
- Reduces confusion and anxiety.
- Encourages focused participation.



FRAME 2: CHUNKING / MEMORIZATION PHASE

Purpose of the Screen

- Acts as the **visual encoding phase** where users observe and memorize items.
- Encourages users to apply chunking strategies naturally.

UI Elements Used

- **Heading:** “REMEMBER THESE!!” – grabs attention and signals urgency.
- **Grid of Icons:**
    - Fruits
    - Animals
    - Musical instruments
- **Countdown Timer (00:05):** Displays limited viewing time.

Chunking Strategy Applied

- Icons are arranged in a **grid format**, making grouping easier.
- Users can mentally chunk items by:
    - Category (fruits, animals, instruments)
    - Visual similarity
- Time restriction forces users to rely on grouping instead of memorizing individually.

UX & Cognitive Benefits

- Enhances short-term memory encoding.
- Encourages pattern recognition.
- Prevents cognitive overload despite multiple items.

FRAME 3: RECALL / SELECTION PHASE

Purpose of the Screen

- Tests how effectively users recall previously seen items.
- Measures the success of chunking.

UI Elements Used

- **Instruction Text:** “CHOOSE THE ITEMS YOU REMEMBER”
- **Grid of Options:**
    - Includes both previously shown items and distractors.
- **Selection Boxes:** Allow users to mark remembered items.
- **SUBMIT Button:** Confirms user choices.

How the Recall Phase Works

1. Users compare current options with their memory.
2. They select items they believe were shown earlier.
3. Distractors test false memory and recall accuracy.

UX & Cognitive Benefits

- Evaluates memory retrieval.
- Encourages careful decision-making.
- Simple interaction reduces task complexity.

FRAME 4: RESULT / FEEDBACK PAGE

Purpose of the Screen

- Displays user performance.
- Provides immediate feedback on recall accuracy.

UI Elements Used

- **Score Message:** “YOU RECALLED 4/6 ITEMS CORRECTLY!”
- **NEXT Button:** Allows progression or continuation.

Feedback Analysis

- Clear numerical feedback helps users understand performance.
- Positive reinforcement motivates continued interaction.

UX & Cognitive Benefits

- Reinforces learning through feedback.
- Makes the experience rewarding rather than stressful.
- Encourages users to retry or improve.

CONCLUSION

This experiment demonstrates that **chunking visual information** significantly improves user memory recall. By grouping related items and limiting exposure time, users are able to remember more efficiently. The UI design minimizes cognitive load while maintaining engagement, proving that chunking is an effective strategy in memory-based user interfaces.

FIGMA
URL:https://www.figma.com/design/WSlk2pQqRhgfANQbobmUHL/EXP2?node-id=1-2&t=TUa3gS9K4v5VZKTh-1